RS05 Motor Notes for Bistable Hook Excitation

Rolly Robot

2026

1 Motor Parameters

Table 1: RS05 parameters used by the bistable simulations.

Parameter	Value	Notes
Rated torque	1.6 N·m	Continuous rating with heat sinking.
Peak torque	5.5 N·m	Short-duration command limit.
No-load output speed	480 RPM ±10%	Equivalent to $\omega_{\max} \approx 50.3\,\mathrm{rad/s}$ at the output shaft.
Torque constant K_t	$0.94\,\mathrm{N{\cdot}m/A_{rms}}$	Used to convert Iq current to torque.
Gear ratio	7.75:1	Planetary reduction.
Drive mode	FOC	Field-oriented control.
Maximum phase current	$11\,\mathrm{A_{pk}}$	The torque command is still limited to the motor's peak torque range.
CAN bus rate	1 Mbps	CAN 2.0B extended frame.
Encoder	14-bit absolute	Used for snap detection and state feedback.

2 Velocity Limit

The 480 RPM value is a shaft angular velocity limit, not a direct frequency limit. For an oscillating shaft command at frequency f and angular amplitude Θ,

$$v_{\mathrm{peak}} = \Theta\, 2\pi f \le \omega_{\max}. \tag{1}$$

Therefore

$$\Theta_{\max}(f) = \frac{\omega_{\max}}{2\pi f}. \tag{2}$$

With a lever arm r, the maximum linear stroke is

$$\delta_{\max}(f) = \frac{\omega_{\max} r}{2\pi f}. \tag{3}$$

3 CAN Control Modes

The RS05 run mode is selected by writing `run_mode` at index `0x7005` with Communication Type 18, then enabling the motor with Type 3.

Table 2: RS05 control modes relevant to sinusoidal excitation.

Value	**Mode**	**Use**
0	Operation	MIT-style torque/position/velocity command. Set $K_p = K_d = 0$ for pure torque feed-forward through t_{ff}.
3	Current	Direct Iq command through `iq_ref`; no position or speed loop.
5	Position CSP	Cyclic synchronous position; useful only if the position trajectory and velocity limit are acceptable.

4 Streaming A Sine Command

The motor firmware does not provide an onboard sine-wave generator. A controller must compute each sample and stream it over CAN.

For operation mode, send Type 1 frames with

$$t_{ff}(t) = \tau_0 \sin(2\pi f t), \qquad K_p = 0, \quad K_d = 0.$$

For current mode, write

$$i_q(t) = \frac{\tau_0}{K_t} \sin(2\pi f t)$$

to `iq_ref` at index `0x7006`. For example, a 2.87 N·m torque amplitude corresponds to about 3.05 A using $K_t = 0.94\,\mathrm{N{\cdot}m/A}$.

5 ESP32 Timing

At 1 Mbps CAN, a 29-bit extended frame is roughly 110 bits, or about 110 μs on the bus. Two motors commanded at 500 Hz each require about 1000 frames/s, so the command traffic is roughly 11% bus utilisation before feedback frames. A 1 ms hardware-timer loop on the ESP32 is adequate if WiFi and Bluetooth are disabled and the CAN/timer work is kept off latency-sensitive tasks.